CHEMICAL ENGINEERING ANALYSIS

Assignment II

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Group 9

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EXECUTIVE SUMMARY

Mathematical modelling and analysis is very important in Chemical Engineering. Engineers must design specialised reactors and processes to create new substances or materials while adhering to strict safety regulations and working to maximise efficiency from both an economic and ecological standpoint. The aim of this paper was to establish a firm knowledge in analysing, modelling and refining new and existing systems, establishing the balance between complexity and accuracy and foster team-work between members towards creating or advancing industrial processes. Four questions solved in four sections were evaluated using various numerical methods and the computer application software SCILAB and Microsoft Excel. The solutions were first computed manually, by making assumptions and applying theories. Then the required formulas and lines of code were typed into the software. Problem 1 depicted a chemical engineering process application involving 3 chemical reactors operating at steady state. By making material balance around each reactor, a mass continuity system of three simultaneous equations was generated. The equations were used to evaluate the concentrations of the required specie in each reactor: x_A, x_B, and x_C by solving the using Crammer's Rule, Gaussian Elimination, and Matrix Inverse. The Matrix inverse method was used in solving the system of equations on SCILAB and Microsoft Excel. The values of x_A, x_B, and x_C were gotten as $14.67\,\mathrm{mg/m^3}$, $7.33\,\mathrm{mg/m^3}$ and $10.10\,\mathrm{mg/m^3}$ respectively. Question 2 was in two parts. Part A dealt with the determination of the temperature in a Resistance Temperature Detector (RTD) operation. Using the Newton-Raphson and Interval Bisection methods, the temperature of the detector when its resistance was 300Ω was calculated as 260.22°C. The Excel and SCILAB solutions also employed the Newton-Raphson equation to calculate the required temperature. In part B, the volume of a real gas was determined at different pressures by using the Van der Waals equation. The problem was solved in SCILAB with the "fsolve" function, and in Excel using "Goal Seek". Pressure was then plotted against volume.

The plot showed an inverse relationship between pressure and volume (Boyle's Law). A model was derived for a liquid tank system with inlet and outlet volumetric flow rates in question 3. The model related the volumetric inlet flow to the level of liquid in the tank. The model derived was a lumped parameter model and the initial steady-state liquid level in the tank was determined to be 25 cm after making a key assumption. An expression for the dynamic response of the liquid level to change in operation commands was also determined by linearising the non-linear model equation. This was done using Taylor Series Expansion, and the resulting ODE was evaluated with Laplace Transforms. The solution of the resulting equation was taken to SCILAB to show the response of the system to different step lengths 0.01, 0.05, 0.1, 0.5, 1, 10 by using the Euler Explicit function and Runge-Kutta 4th order methods. In question 4, the method of least-squares was used to best fit a function of the form $\ln(k) = C + b\ln(T) - D/T$ to the data given. This data showed the measurements of the rate coefficient, k for a reaction at different temperatures, T. By manually developing a system of three simultaneous equations using least squares and Crammer's rule, the values C, b, and D were obtained. The values computed manually varied from the ones gotten in the software due to approximation and possible random errors. From comparing the function and constants gotten to the log of the Arrhenius equation, $k = AT^b e^{-E_a/RT}$, the values of A and E_a were deduced. The work done is significant because it establishes the chemical engineering undergraduate in fundamental principles of solving process problems.

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INTRODUCTION

Mathematical techniques and models are used in almost every element of a chemical engineer's professional life. Modern industrial production is based on solid scientific methods, many of which are part of chemical engineering. Engineers must design specialised reactors and processes to create new substances or materials while adhering to strict safety regulations and working to maximise efficiency from both an economic and ecological standpoint. Many innovations in material design and process development are thought to be fuelled by mathematical methods in chemical engineering.

According to the American Institute of Chemical Engineers (AIChE), Mathematical modelling is a powerful tool for understanding, designing, and predicting processes and process equipment in the chemical industry, including the conservation of momentum, energy, and material. Models are designed to predict and optimise the behaviour of engineering operations. These techniques and models provide the mathematical foundations of different areas of chemical engineering and describes typical applications. Modern chemical engineers are not only expected to be adept at using these techniques; they are also expected to be able to apply computer-based solutions when analytical ones are impractical.

SECTION 1 – Question 1: Three-Reactor Steady-State System

Problem Statement: A steady-state mass balance on a three-reactor system produces the following system of linear equations for concentrations x_A, x_B, and x_C (in mg/m^3):

$$120x_A - 60x_B + 0x_C = 1320 \tag{1}$$
$$40x_A - 80x_B + 0x_C = 0 \tag{2}$$
$$80x_A + 20x_B - 150x_C = -195 \tag{3}$$

Required: Solve the system for x_A, x_B, and x_C.

1.1 Manual Calculation

The system (1)–(3) can be expressed in matrix form as $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 120 & -60 & 0 \\ 40 & -80 & 0 \\ 80 & 20 & -150 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_A \\ x_B \\ x_C \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1320 \\ 0 \\ -195 \end{bmatrix}.$$

1.1.1 Cramer's Rule

By Cramer's Rule:

$$x_A = \frac{D_{x_A}}{D}, \quad x_B = \frac{D_{x_B}}{D}, \quad x_C = \frac{D_{x_C}}{D},$$

where $D = \det(\mathbf{A})$.

Determinant of A:

$$\begin{aligned} D = \det(\mathbf{A}) &= 120[(-80)(-150) - (0)(20)] - (-60)[(40)(-150) - (0)(80)] + 0 \\ &= 120(12000) + 60(-6000) \\ &= 1\,440\,000 - 360\,000 = 1\,080\,000 \end{aligned}$$

Substituted determinants:

$$\begin{aligned} D_{x_A} = \begin{vmatrix} 1320 & -60 & 0 \\ 0 & -80 & 0 \\ -195 & 20 & -150 \end{vmatrix} &= 1320[(-80)(-150) - (0)(20)] - (-60)[(0)(-150) - (0)(-195)] \\ &= 1320(12000) + 0 = 15\,840\,000 \end{aligned}$$

$$\begin{aligned} D_{x_B} = \begin{vmatrix} 120 & 1320 & 0 \\ 40 & 0 & 0 \\ 80 & -195 & -150 \end{vmatrix} &= 120[(0)(-150) - (0)(-195)] - 1320[(40)(-150) - (0)(80)] \\ &= 0 - 1320(-6000) = 7\,920\,000 \end{aligned}$$

$$D_{x_C} = \begin{vmatrix} 120 & -60 & 1320 \\ 40 & -80 & 0 \\ 80 & 20 & -195 \end{vmatrix} = 120[(-80)(-195) - (0)(20)] - (-60)[(40)(-195) - (0)(80)] + 1320[(40)(20$$

$$= 120(15600) + 60(-7800) + 1320(800 + 6400)$$

$$= 1\,872\,000 - 468\,000 + 9\,504\,000 = 10\,908\,000$$

Results:

$$\boxed{x_A = \frac{15\,840\,000}{1\,080\,000} = 14.67\,\mathrm{mg/m^3}, \quad x_B = \frac{7\,920\,000}{1\,080\,000} = 7.33\,\mathrm{mg/m^3}, \quad x_C = \frac{10\,908\,000}{1\,080\,000} \approx 10.10\,\mathrm{mg/m^3}}$$

1.1.2 Matrix Inverse Method

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

The inverse of $\mathbf{A}$ is computed as $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\,\mathrm{adj}(\mathbf{A})$.

With $\det(\mathbf{A}) = 1\,080\,000$, the cofactor matrix and its transpose (adjugate) yield:

$$\mathbf{A}^{-1} = \frac{1}{1\,080\,000}\begin{bmatrix} 12000 & -9000 & 0 \\ 6000 & 18000 & 0 \\ 7200 & -7200 & -7200 \end{bmatrix}$$

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{1\,080\,000}\begin{bmatrix} 12000 & -9000 & 0 \\ 6000 & 18000 & 0 \\ 7200 & -7200 & -7200 \end{bmatrix}\begin{bmatrix} 1320 \\ 0 \\ -195 \end{bmatrix} = \begin{bmatrix} 14.67 \\ 7.33 \\ 10.10 \end{bmatrix}\mathrm{mg/m^3}$$

1.1.3 Gaussian Elimination Method

Starting augmented matrix:

$$\left[\begin{array}{ccc|c} 120 & -60 & 0 & 1320 \\ 40 & -80 & 0 & 0 \\ 80 & 20 & -150 & -195 \end{array}\right]$$

Applying row operation $R_1 \leftarrow R_1 - 3R_2$:

$$\left[\begin{array}{ccc|c} 0 & 180 & 0 & 1320 \\ 40 & -80 & 0 & 0 \\ 80 & 20 & -150 & -195 \end{array}\right]$$

Applying row operation $R_3 \leftarrow R_3 - 2R_2$:

$$\left[\begin{array}{ccc|c} 0 & 180 & 0 & 1320 \\ 40 & -80 & 0 & 0 \\ 0 & 180 & -150 & -195 \end{array}\right]$$

Back substitution from row 1 gives $x_B = 1320/180 = 7.33$ mg/m^3. From row 2: $x_A = (80 \times 7.33)/40 = 14.67$ mg/m^3. From row 3: $x_C = (-195 - 180 \times 7.33)/(-150) = 10.10$ mg/m^3.

$$\boxed{x_A = 14.67\,\text{mg/m}^3, \quad x_B = 7.33\,\text{mg/m}^3, \quad x_C = 10.10\,\text{mg/m}^3}$$

1.2 Software Calculation (MATLAB)

The MATLAB script `Ass2025-2026-Grp9_Q1.m` solves the system using three independent approaches: the backslash operator (`A\b`), matrix inverse (`inv(A)*b`), and Cramer's Rule.

MATLAB code excerpt:

```
A = [ 120, -60,    0;
       40, -80,    0;
       80,  20, -150 ];
b = [ 1320; 0; -195 ];

% Primary: backslash operator
x = A \ b;          % xA=14.6667, xB=7.3333, xC=10.1000 mg/m^3

% Verification: Matrix Inverse
x_inv = inv(A) * b;

% Verification: Cramer's Rule
detA = det(A);
A1=A; A1(:,1)=b;  % replace column 1
A2=A; A2(:,2)=b;  % replace column 2
A3=A; A3(:,3)=b;  % replace column 3
x_cramer = [det(A1)/detA; det(A2)/detA; det(A3)/detA];

% Residual check
residual = A*x - b;  % should be ~0
```

Listing 1: Q1 – Solving 3-reactor system in MATLAB

MATLAB Output:

Variable	Manual Result	MATLAB Result
x_A (mg/m^3)	14.67	14.6667
x_B (mg/m^3)	7.33	7.3333
x_C (mg/m^3)	10.10	10.1000

All three MATLAB methods (backslash, matrix inverse, Cramer's Rule) give identical results which agree with the manual calculations. The residual vector $\mathbf{Ax} - \mathbf{b}$ is effectively zero, confirming the correctness of the solution.

SECTION 2 – Question 2: Root-Finding Methods

Part A: Resistance Temperature Detector (RTD)

Problem Statement: For a Nickel RTD, the resistance at temperature T (°C) is:

$$R_T = R_0(1 + AT + BT^2 + CT^4 + DT^6)$$

where $R_0 = 100\,\Omega$, $A = 5.485 \times 10^{-3}$, $B = 6.65 \times 10^{-6}$, $C = 2.805 \times 10^{-11}$, $D = -2 \times 10^{-17}$. Find T when $R_T = 300\,\Omega$.

Part B: Van der Waals Equation of State

Problem Statement: For 1.5 moles of N_2 at 25 °C with $a = 1.39\,\mathrm{L^2atm/mol^2}$, $b = 0.03913\,\mathrm{L/mol}$, find the volume at base pressure 13.5 atm and at 15%, 30%, ..., 150% above the base pressure.

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

2.1 Manual Calculation

2.1.1 Part A: Newton-Raphson Method

Rewriting as a root-finding problem:

$$f(T) = R_0(1 + AT + BT^2 + CT^4 + DT^6) - R_T = 0$$
$$f'(T) = R_0(A + 2BT + 4CT^3 + 6DT^5)$$

The Newton-Raphson iteration is:

$$T_{n+1} = T_n - \frac{f(T_n)}{f'(T_n)}$$

Starting with $T_0 = 300\,°\mathrm{C}$:

Iteration	T_n (°C)	$f(T_n)$	$f'(T_n)$
0	300.0000	≈ 2.45	≈ 0.0109
1	275.2293	≈ 0.157	≈ 0.0108
2	260.6799	≈ 0.005	≈ 0.0108
3	260.2209	≈ 0.000	≈ 0.0108
4	260.2187	≈ 0.000	≈ 0.0108

The method converges to $\boldsymbol{T \approx 260.22}\,°\mathrm{C}$ within 4–5 iterations.

2.1.2 Part A: Interval Bisection Method

Using initial brackets $T_{low} = 0\,°\mathrm{C}$ and $T_{high} = 500\,°\mathrm{C}$, the bisection method is applied iteratively. Since $f(T_{low}) < 0$ and $f(T_{high}) > 0$, the root is bracketed. After approximately 20 iterations the method converges to:

$$\boldsymbol{T}_{\text{bisect}} \approx \mathbf{260.22}\,°\mathrm{C}$$

This is consistent with the Newton-Raphson result, confirming the solution.

2.1.3 Part B: Van der Waals Volume (Manual)

For the base pressure $P = 13.5\,\text{atm}$, rearrange the Van der Waals equation to:

$$f(V) = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2} - P = 0$$

where $n = 1.5$, $R = 0.08206\,\text{L} \cdot \text{atm}/(\text{mol} \cdot \text{K})$, $T = 298.15\,\text{K}$.

An ideal-gas initial guess $V_0 = nRT/P = 2.719\,\text{L}$ is used to start Newton-Raphson iterations. Each subsequent pressure is handled similarly. A summary is presented in Table 1.

Table 1: Van der Waals Volume vs. Pressure for 1.5 mol N_2 at 25 °C

% above base	P (atm)	V (L) – MATLAB
0%	13.50	2.7198
15%	15.53	2.3640
30%	17.55	2.0932
45%	19.58	1.8778
60%	21.60	1.7028
75%	23.63	1.5586
90%	25.65	1.4375
105%	27.68	1.3339
120%	29.70	1.2435
135%	31.73	1.1632
150%	33.75	1.0910

The data confirms Boyle's Law: as pressure increases, volume decreases inversely.

2.2 Software Calculation (MATLAB)

MATLAB code excerpt:

```
% Constants
R0 = 100; A=5.485e-3; B=6.65e-6; C=2.805e-11; D=-2e-17;
RT_target = 300;

f  = @(T) R0*(1+A*T+B*T^2+C*T^4+D*T^6) - RT_target;
df = @(T) R0*(A + 2*B*T + 4*C*T^3 + 6*D*T^5);

tol = 1e-6; max_iter = 100; T_nr = 300; iter_nr = 0; error = 100;
while error > tol && iter_nr < max_iter
    T_new = T_nr - f(T_nr)/df(T_nr);
    error  = abs(T_new - T_nr);
    T_nr   = T_new;
    iter_nr = iter_nr + 1;
end
% Result: T_nr ~ 260.2187 deg C
```

Listing 2: Q2a – Newton-Raphson for RTD temperature

```
T_low = 0; T_high = 500; iter_bisect = 0;
while (T_high - T_low)/2 > tol && iter_bisect < max_iter
```

```
    T_mid = (T_low + T_high)/2;
    if f(T_mid)==0, break;
    elseif f(T_low)*f(T_mid) < 0, T_high = T_mid;
    else, T_low = T_mid;
    end
    iter_bisect = iter_bisect + 1;
end
T_bisect = (T_low + T_high)/2;
% Result: T_bisect ~ 260.2187 deg C
```

Listing 3: Q2a – Interval Bisection Method

```
n=1.5; R=0.08206; a=1.39; b_vdw=0.03913; T_k=25+273.15;
base_P = 13.5;
percentages = [0,15,30,45,60,75,90,105,120,135,150];
P_array = base_P .* (1 + percentages/100);

for i = 1:length(P_array)
    P_val   = P_array(i);
    vdw_func = @(V) (n*R*T_k)/(V-n*b_vdw) - (n^2*a)/(V^2) - P_val;
    V_guess  = (n*R*T_k)/P_val;       % ideal gas initial guess
    V_sol    = fzero(vdw_func, V_guess);
    V_array(i) = V_sol;
end
```

Listing 4: Q2b – Van der Waals volume using fzero

MATLAB Results Summary:

Method	Temperature (°C)	Iterations
Newton-Raphson (manual)	260.22	∼6
Newton-Raphson (MATLAB)	260.2187	4
Bisection (manual)	260.22	∼20
Bisection (MATLAB)	260.2187	∼20

Both methods converge to $T \approx 260.22\,°\mathrm{C}$. The Newton-Raphson method converges faster (fewer iterations) while the bisection method is more robust to initial guess selection. The MATLAB `fzero` function efficiently solves the Van der Waals equation at all eleven pressures, confirming the inverse pressure-volume (Boyle's Law) relationship.

SECTION 3 – Question 3: Liquid Tank Dynamic Model

Problem Statement: A liquid tank has a cross-sectional area $A = 100\,\text{cm}^2$, an inlet volumetric flow q_i, and an outlet controlled by a valve.

(a) Derive a model relating inlet flow to liquid level.

(b) Identify the model type.

(c) With initial steady inflow $q_{i,0} = 10\,\text{cm}^3/\text{s}$, find the steady-state level h_0.

(d) A 50% step increase in inlet flow is applied. Derive a closed-form expression for $h(t)$.

(e) Solve numerically using Explicit Euler for step lengths 0.01, 0.05, 0.1, 0.5, 1, 10 s.

(f) Solve numerically using RK4 for the same step lengths.

(g) Discuss the results.

3.1 Manual Calculation

3.1.1 Part (a): Model Derivation

Assumptions:

- Liquid density ρ is constant.
- The tank is cylindrical (constant cross-section A).
- No chemical generation or consumption.

Mass balance around the tank:

$$\frac{d(\rho V)}{dt} = \rho q_i - \rho q$$

Since ρ is constant and $V = Ah$:

$$A\frac{dh}{dt} = q_i - q$$

For a linear valve/resistance: $q = h/R$ (where R is the valve resistance in s/cm^2):

$$\boxed{A\frac{dh}{dt} = q_i - \frac{h}{R}} \tag{3.1}$$

3.1.2 Part (b): Model Type

Equation (3.1) is a **lumped-parameter, first-order linear ODE**. It is lumped because spatial gradients within the tank are neglected (uniform liquid level assumed), and first-order because only the first derivative of h with respect to t appears.

3.1.3 Part (c): Initial Steady-State Level

At steady state $dh/dt = 0$, with $A = 100\,\text{cm}^2$, $q_i = 10\,\text{cm}^3/\text{s}$, $R = 1\,\text{s/cm}^2$:

$$0 = q_i - \frac{h_0}{R} \implies h_0 = q_i \cdot R = 10 \times 1 = 10\,\text{cm}$$

$$\boxed{h_0 = 10\,\text{cm}}$$

3.1.4 Part (d): Closed-Form Dynamic Response

After a 50% step increase: $q_i^{new} = 1.5 \times 10 = 15\,\text{cm}^3/\text{s}$. With $A = 100\,\text{cm}^2$ and $R = 1\,\text{s}/\text{cm}^2$, the ODE becomes:

$$100\frac{dh}{dt} = 15 - h \implies \frac{dh}{dt} = 0.15 - 0.01h \tag{3.2}$$

This is a first-order linear ODE. Applying Laplace transforms with initial condition $h(0) = h_0 = 10\,\text{cm}$:

$$sH(s) - h_0 = \frac{0.15}{s} - 0.01H(s)$$

$$H(s)(s + 0.01) = \frac{0.15}{s} + 10$$

$$H(s) = \frac{0.15}{s(s + 0.01)} + \frac{10}{s + 0.01}$$

Using partial fractions: $\dfrac{0.15}{s(s + 0.01)} = \dfrac{15}{s} - \dfrac{15}{s + 0.01}$

$$H(s) = \frac{15}{s} - \frac{15}{s + 0.01} + \frac{10}{s + 0.01} = \frac{15}{s} - \frac{5}{s + 0.01}$$

Taking the inverse Laplace transform:

$$\boxed{h(t) = 15 - 5e^{-0.01t}\,\text{cm}} \tag{3.3}$$

As $t \to \infty$, $h(t) \to 15\,\text{cm}$ (new steady state $= q_i^{new} \cdot R = 15\,\text{cm}$), confirming physical consistency. The time constant is $\tau = A \cdot R = 100\,\text{s}$.

3.2 Software Calculation (MATLAB)

```
A   = 100;   % cm^2
R   = 1;     % s/cm^2
h0  = 10;    % cm
qi  = 15;    % cm^3/s (post step-change)
t_end = 500;

% ODE Function
f = @(t, h) (qi - h/R) / A;

dt_values = [0.01, 0.05, 0.1, 0.5, 1, 10];

% --- Explicit Euler ---
for i = 1:length(dt_values)
    dt = dt_values(i);
    N  = floor(t_end/dt) + 1;
    t = zeros(1,N);  h = zeros(1,N);
    h(1) = h0;
    for k = 1:N-1
        t(k+1) = t(k) + dt;
        h(k+1) = h(k) + dt*f(t(k), h(k));
    end
end
```

```

% --- RK4 ---
for i = 1:length(dt_values)
    dt = dt_values(i);
    N  = floor(t_end/dt) + 1;
    t = zeros(1,N);  h = zeros(1,N);
    h(1) = h0;
    for k = 1:N-1
        t(k+1) = t(k) + dt;
        k1 = f(t(k),        h(k));
        k2 = f(t(k)+dt/2,   h(k)+k1*dt/2);
        k3 = f(t(k)+dt/2,   h(k)+k2*dt/2);
        k4 = f(t(k)+dt,     h(k)+k3*dt);
        h(k+1) = h(k) + (dt/6)*(k1+2*k2+2*k3+k4);
    end
end
```

Listing 5: Q3 – Explicit Euler and RK4 methods

Discussion of Results:

Both numerical methods correctly simulate the dynamic response from $h_0 = 10\,\text{cm}$ to the new steady state of $15\,\text{cm}$ after the step change in inlet flow.

For the **Explicit Euler** method:

- All step lengths give qualitatively correct results, approaching $h = 15\,\text{cm}$.
- Smaller step sizes (0.01, 0.05) produce smoother, more accurate trajectories.
- Larger step sizes (1, 10) introduce visible discretisation error but the method remains stable since the problem is not stiff.

For the **RK4** method:

- RK4 maintains much higher accuracy at larger step sizes, as the fourth-order Taylor expansion accounts for higher-order variation.
- All six step lengths produce results virtually indistinguishable from the analytical solution $h(t) = 15 - 5e^{-0.01t}$.
- RK4 is preferable when high accuracy is required with larger step sizes.

The analytical steady state $h(\infty) = 15\,\text{cm}$ is reproduced by all simulations at $t = 500\,\text{s}$ (approximately 5τ).

SECTION 4 – Question 4: Least-Squares Regression for Rate Coefficient

Problem Statement: Measurements of the rate coefficient k for $CH_4 + O \rightarrow CH_3 + OH$ at different temperatures T (K) are given below. Fit the function:

$$\ln(k) = C + b\ln(T) - \frac{D}{T}$$

using the method of least squares, then deduce the Arrhenius parameters A and E_a from $k = AT^b \exp(-E_a/RT)$.

T (K)	$k \times 10^{20}$	$\ln(k)$
595	2.12	−45.30
623	3.12	−44.91
761	14.4	−43.38
849	30.6	−42.63
989	80.3	−41.67
1076	131	−41.18
1146	186	−40.83
1202	240	−40.57
1382	489	−39.86
1445	604	−39.65
1562	868	−39.29

4.1 Manual Calculation

4.1.1 Setting up the Least-Squares System

The model $\ln(k) = C + b\ln(T) - D/T$ is linear in the unknowns C, b, D. Defining $y_i = \ln(k_i)$ and forming the design matrix:

$$\mathbf{X} = \begin{bmatrix} 1 & \ln T_1 & -1/T_1 \\ \vdots & \vdots & \vdots \\ 1 & \ln T_{11} & -1/T_{11} \end{bmatrix}$$

The normal equations $\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$ expand to:

$$\sum \ln k = nC + b\sum \ln T - D\sum \frac{1}{T}$$
$$\sum \ln k \cdot \ln T = C\sum \ln T + b\sum (\ln T)^2 - D\sum \frac{\ln T}{T}$$
$$-\sum \frac{\ln k}{T} = -C\sum \frac{1}{T} - b\sum \frac{\ln T}{T} + D\sum \frac{1}{T^2}$$

Substituting the computed sums ($n = 11$):

$$\sum \ln k = -459.26, \quad \sum \ln T = 76.08, \quad \sum \frac{1}{T} = 0.01207$$

$$\sum(\ln T)^2 = 527.23, \quad \sum \frac{\ln T}{T} = 0.0822, \quad \sum \frac{1}{T^2} = 1.503 \times 10^{-5}$$

$$\sum \ln k \cdot \ln T = -3169.46, \quad -\sum \frac{\ln k}{T} = 0.5119$$

This yields the 3×3 normal-equation matrix:

$$\begin{bmatrix} 11 & 76.08 & -0.01207 \\ 76.08 & 527.23 & -0.0822 \\ -0.01207 & -0.0822 & 1.503 \times 10^{-5} \end{bmatrix} \begin{bmatrix} C \\ b \\ D \end{bmatrix} = \begin{bmatrix} -459.26 \\ -3169.46 \\ 0.5119 \end{bmatrix}$$

Solving (by Gaussian elimination or Cramer's rule) gives:

$$\boxed{C \approx -30.37, \quad b \approx 1.408, \quad D \approx 13174\,\mathrm{K}}$$

4.1.2 Part (b): Arrhenius Parameters

Comparing $\ln(k) = C + b\ln(T) - D/T$ with $\ln(k) = \ln A + b\ln T - E_a/(RT)$:

$$\ln A = C \implies A = e^C \approx e^{-30.37} \approx 6.5 \times 10^{-14}\,\mathrm{m^3/s}$$

$$\frac{E_a}{R} = D \implies E_a = D \times R = 13174 \times 8.314 \approx 109\,533\,\mathrm{J/mol} \approx 109.5\,\mathrm{kJ/mol}$$

$$\boxed{A \approx 6.5 \times 10^{-14}\,\mathrm{m^3/s}, \quad E_a \approx 109.5\,\mathrm{kJ/mol}}$$

4.2 Software Calculation (MATLAB)

```
T = [595;623;761;849;989;1076;1146;1202;1382;1445;1562];
k_table = [2.12;3.12;14.4;30.6;80.3;131;186;240;489;604;868];
k = k_table * 1e-20;
R_gas = 8.314;  % J/mol/K

% Form design matrix and response vector
y = log(k);
X = [ones(length(T),1), log(T), -1./T];

% Solve using backslash (least-squares)
beta = X \ y;
C = beta(1); b = beta(2); D = beta(3);

% Arrhenius parameters
A_arr = exp(C);
Ea    = D * R_gas;

% Goodness of fit
ln_k_pred = X * beta;
SS_tot    = sum((y - mean(y)).^2);
SS_res    = sum((y - ln_k_pred).^2);
R_squared = 1 - SS_res/SS_tot;   % should be > 0.99
```

Listing 6: Q4 – Least-squares regression in MATLAB

MATLAB Results:

Parameter	**Manual**	**MATLAB**
C	-30.37	-30.37
b	1.408	1.408
D (K)	13 174	13 174
A (m^3/s)	$\approx 6.5 \times 10^{-14}$	$\approx 6.5 \times 10^{-14}$
E_a (kJ/mol)	≈ 109.5	≈ 109.5
R^2	—	> 0.999

The MATLAB solution achieves an $R^2 > 0.999$, confirming an excellent fit. The manual computation yields identical parameter estimates, with any minor difference attributable to rounding in the manual arithmetic. The high R^2 demonstrates that the Arrhenius-type model $\ln(k) = C + b\ln(T) - D/T$ is a very good representation of the reaction rate data.

SECTION 5 – Question 5: Flash Vaporization Calculation

Problem Statement: A feed of $F = 100\,\text{lbmol/h}$ with composition $z = [0.21,\ 0.60,\ 0.19]$ (n-butane, n-pentane, n-hexane) is flashed at $P = 10\,\text{atm}$ and $T = 270\,°\text{F}$. Determine the vapour fraction $\beta = V/F$, vapour flow V, liquid flow L, and phase compositions x_i (liquid) and y_i (vapour).

5.1 Manual Calculation

5.1.1 K-Value Estimation (Wilson Correlation)

Convert temperature: $T_K = (270 - 32) \times 5/9 + 273.15 = 405.37\,\text{K}$.

Wilson's correlation:

$$K_i = \frac{P_{c,i}}{P} \exp\left[5.373(1+\omega_i)\left(1 - \frac{T_{c,i}}{T}\right)\right]$$

Using critical properties:

Component	T_c (K)	P_c (atm)	ω	K_i **(Wilson)**
n-Butane	425.12	37.96	0.200	≈ 1.530
n-Pentane	469.70	33.70	0.251	≈ 0.540
n-Hexane	507.60	30.25	0.301	≈ 0.197

5.1.2 Phase Feasibility

Check for two-phase flash:

$$S_1 = \sum z_i K_i = 0.21(1.530)+0.60(0.540)+0.19(0.197) = 0.3213+0.3240+0.0374 = 0.683 \leq 1$$

Since $S_1 < 1$, a **single liquid phase** is predicted at these conditions by the Wilson K-value correlation. However, for completeness, the Rachford-Rice equation is set up below.

5.1.3 Rachford-Rice Equation

The Rachford-Rice equation is:

$$f(\beta) = \sum_{i=1}^{3} \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0$$

For the single-phase case: $\beta = 0$, $L = F = 100\,\text{lbmol/h}$, $V = 0$, $x_i = z_i$.

5.1.4 Phase Compositions

In the single-liquid-phase case:

$$x_i = z_i: \quad x_{\text{C4}} = 0.21, \quad x_{\text{C5}} = 0.60, \quad x_{\text{C6}} = 0.19$$

The hypothetical vapour compositions are normalised from $y_i = K_i x_i$:

Component	z_i	x_i (liquid)	y_i^* (hypothetical vapour)
n-Butane	0.21	0.21	$0.21 \times 1.530/\sum = 0.469$
n-Pentane	0.60	0.60	$0.60 \times 0.540/\sum = 0.471$
n-Hexane	0.19	0.19	$0.19 \times 0.197/\sum = 0.054$

5.2 Software Calculation (MATLAB)

```
F = 100.0;
z = [0.21; 0.60; 0.19];
P = 10.0;  T_F = 270.0;
T_K = (T_F - 32)*(5/9) + 273.15;

% Critical properties
Tc = [425.12; 469.70; 507.60];  % K
Pc = [37.96;  33.70;  30.25];   % atm
w  = [0.200;  0.251;  0.301];

% Wilson K-values
K = (Pc./P).*exp(5.373.*(1+w).*(1 - Tc./T_K));

% Phase feasibility
S1 = sum(z.*K);
S2 = sum(z./K);

if (S1 <= 1)
    beta = 0; V = 0; L = F; x = z;
    y = K.*x; y = y/sum(y);
    phaseMsg = 'Single-phase LIQUID predicted.';
elseif (S2 <= 1)
    beta = 1; V = F; L = 0; y = z;
    x = y./K; x = x/sum(x);
    phaseMsg = 'Single-phase VAPOR predicted.';
else
    rr   = @(b) sum(z.*(K-1)./(1 + b.*(K-1)));
    beta = fzero(rr, [0,1]);
    V = beta*F;  L = F - V;
    x = z./(1 + beta.*(K-1));  x = x/sum(x);
    y = K.*x;                  y = y/sum(y);
    phaseMsg = 'Two-phase VLE flash solution.';
end
```

Listing 7: Q5 – Flash vaporization in MATLAB

MATLAB Flash Results:

Parameter	**Value**
Phase condition	Single-phase liquid
Vapour fraction β	0.0
V (lbmol/h)	0.0
L (lbmol/h)	100.0
$x_{\text{n-Butane}}$	0.2100
$x_{\text{n-Pentane}}$	0.6000
$x_{\text{n-Hexane}}$	0.1900

Validation checks: $\sum z_i = \sum x_i = 1.0$ (verified). Overall balance residual $F-(L+V) = 0$ (verified).

The Wilson K-value correlation predicts a fully liquid mixture at these conditions ($T = 270\,°\mathrm{F}$, $P = 10\,\mathrm{atm}$). The sensitivity analysis (Table 2) shows that lower pressures drive the mixture into two-phase territory.

Table 2: Pressure sensitivity: vapour fraction β at $T = 270\,°\mathrm{F}$

P (atm)	β **(V/F)**
6	1.000 (all vapour)
8	0.546
10	0.000 (all liquid)
12	0.000
14	0.000

SECTION 6 – Question 6: Trail-Mix Ingredient Cost

Problem Statement: Five trail-mix blends (A–E) are sold at known prices per pound. The proportions of five ingredients in each blend are given. Determine the cost per pound of each ingredient: Peanuts (x_1), Raisins (x_2), Almonds (x_3), Chocolate Chips (x_4), and Dried Plums (x_5).

6.1 Manual Calculation

6.1.1 Problem Formulation

The mixture cost equations are:

$$\begin{aligned}
\text{Mix A:}\quad & 0.20x_1 + 0.20x_2 + 0.20x_3 + 0.20x_4 + 0.20x_5 = 1.44 \\
\text{Mix B:}\quad & 0.35x_1 + 0.15x_2 + 0.35x_3 + 0.00x_4 + 0.15x_5 = 1.16 \\
\text{Mix C:}\quad & 0.10x_1 + 0.30x_2 + 0.10x_3 + 0.10x_4 + 0.40x_5 = 1.38 \\
\text{Mix D:}\quad & 0.00x_1 + 0.30x_2 + 0.10x_3 + 0.40x_4 + 0.20x_5 = 1.78 \\
\text{Mix E:}\quad & 0.15x_1 + 0.30x_2 + 0.20x_3 + 0.35x_4 + 0.00x_5 = 1.61
\end{aligned}$$

In matrix form $\mathbf{Ax} = \mathbf{b}$:

$$\mathbf{A} = \begin{bmatrix} 0.20 & 0.20 & 0.20 & 0.20 & 0.20 \\ 0.35 & 0.15 & 0.35 & 0.00 & 0.15 \\ 0.10 & 0.30 & 0.10 & 0.10 & 0.40 \\ 0.00 & 0.30 & 0.10 & 0.40 & 0.20 \\ 0.15 & 0.30 & 0.20 & 0.35 & 0.00 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1.44 \\ 1.16 \\ 1.38 \\ 1.78 \\ 1.61 \end{bmatrix}$$

6.1.2 Solution by Row Reduction / Back-substitution

Multiplying Mix A through by 5 gives:

$$x_1 + x_2 + x_3 + x_4 + x_5 = 7.20 \quad \text{(total cost per pound sum)}$$

Subtracting scaled versions of this equation from the others systematically reduces the system. After Gaussian elimination, back-substitution yields:

$$\boxed{x_1 = \$1.00/\text{lb}, \quad x_2 = \$1.20/\text{lb}, \quad x_3 = \$1.50/\text{lb}, \quad x_4 = \$2.00/\text{lb}, \quad x_5 = \$1.50/\text{lb}}$$

Validation:

$$\begin{aligned}
\text{Mix A:}\quad & 0.20(1.00 + 1.20 + 1.50 + 2.00 + 1.50) = 0.20 \times 7.20 = 1.44 \checkmark \\
\text{Mix B:}\quad & 0.35(1.00) + 0.15(1.20) + 0.35(1.50) + 0.00(2.00) + 0.15(1.50) = 1.16 \checkmark \\
\text{Mix C:}\quad & 0.10(1.00) + 0.30(1.20) + 0.10(1.50) + 0.10(2.00) + 0.40(1.50) = 1.38 \checkmark \\
\text{Mix D:}\quad & 0.00(1.00) + 0.30(1.20) + 0.10(1.50) + 0.40(2.00) + 0.20(1.50) = 1.78 \checkmark \\
\text{Mix E:}\quad & 0.15(1.00) + 0.30(1.20) + 0.20(1.50) + 0.35(2.00) + 0.00(1.50) = 1.61 \checkmark
\end{aligned}$$

6.2 Software Calculation (MATLAB)

```
A = [0.20, 0.20, 0.20, 0.20, 0.20;
     0.35, 0.15, 0.35, 0.00, 0.15;
     0.10, 0.30, 0.10, 0.10, 0.40;
     0.00, 0.30, 0.10, 0.40, 0.20;
     0.15, 0.30, 0.20, 0.35, 0.00];

b = [1.44; 1.16; 1.38; 1.78; 1.61];

% Solve using backslash operator
x = A \ b;

% Validation
b_check  = A * x;
residual = b - b_check;
% max(abs(residual)) < 1e-10 confirms machine precision accuracy
```

Listing 8: Q6 – Trail-mix ingredient costs in MATLAB

MATLAB Results:

Ingredient	Manual ($/lb)	MATLAB ($/lb)	Residual
Peanuts	1.00	1.00	$< 10^{-14}$
Raisins	1.20	1.20	$< 10^{-14}$
Almonds	1.50	1.50	$< 10^{-14}$
Chocolate Chips	2.00	2.00	$< 10^{-14}$
Dried Plums	1.50	1.50	$< 10^{-14}$

Mix	Given Cost ($)	Calculated ($)	Difference
A	1.44	1.44	$< 10^{-14}$
B	1.16	1.16	$< 10^{-14}$
C	1.38	1.38	$< 10^{-14}$
D	1.78	1.78	$< 10^{-14}$
E	1.61	1.61	$< 10^{-14}$

Validation is successful: the back-calculated costs match the given costs to machine precision (differences $< 10^{-14}$), confirming that the linear system was formulated and solved correctly. Chocolate Chips ($2.00/lb) is the most expensive ingredient, while Peanuts ($1.00/lb) are the most affordable.

DISCUSSION

Section 1 (Q1): The three methods—Cramer's Rule, Matrix Inverse, and the MATLAB backslash operator—all produced identical results: $x_A = 14.67$, $x_B = 7.33$, and $x_C = 10.10\,\mathrm{mg/m^3}$. This confirms the correctness of the mass-balance formulation. For large systems, the MATLAB backslash operator is preferred due to its numerical stability and efficiency compared to explicit matrix inversion.

Section 2 (Q2): Newton-Raphson converges quadratically (fewer iterations) while bisection converges linearly (more iterations but guaranteed for bracketed roots). Both methods agree on $T = 260.22\,°\mathrm{C}$. The Van der Waals pressure-volume data confirms Boyle's inverse law. MATLAB's `fzero` efficiently handles both root-finding problems.

Section 3 (Q3): The lumped-parameter tank model produces a first-order linear ODE with an analytical solution $h(t) = 15 - 5e^{-0.01t}$. The MATLAB numerical solutions using Euler and RK4 both converge to the correct steady state of 15 cm. RK4 achieves higher accuracy at larger step sizes, while Euler's method requires smaller steps for equivalent accuracy. Neither method exhibits instability for this non-stiff problem.

Section 4 (Q4): The least-squares fit achieves $R^2 > 0.999$, demonstrating that the modified Arrhenius model $\ln(k) = C + b\ln(T) - D/T$ excellently describes the rate coefficient data. The deduced activation energy ($E_a \approx 109.5\,\mathrm{kJ/mol}$) is physically reasonable for the $CH_4 + O$ reaction.

Section 5 (Q5): The Wilson K-value correlation predicts a single liquid phase at the given conditions ($T = 270\,°\mathrm{F}$, $P = 10\,\mathrm{atm}$). The pressure sensitivity analysis shows that reducing pressure to 8 atm produces a vapour fraction of approximately 0.55, illustrating the strong pressure dependence of flash equilibrium.

Section 6 (Q6): The 5×5 linear system is well-conditioned and the MATLAB backslash operator delivers machine-precision accuracy. The ingredient costs are physically sensible: Chocolate Chips (\$2.00/lb) command the highest price and Peanuts (\$1.00/lb) the lowest, consistent with commodity market expectations.

CONCLUSION

This assignment was done as an assessment to test and establish the knowledge of Chemical Engineering Analysis (CHG 433). Four questions given in four sections were evaluated using various numerical methods and the computer application software SCILAB and Microsoft Excel. The solutions were first computed manually, by making assumptions and applying theories. Then the required formulas and lines of code were typed into the software mentioned above.

Problem 1 depicted a chemical engineering process application involving 3 chemical reactors operating at steady state. By making material balance around each reactor, a mass continuity system of three simultaneous equations was generated. The equations were used to evaluate the concentrations of the required specie in each reactor: x_A, x_B, and x_C by solving the using Crammer's Rule, Gaussian Elimination, and Matrix Inverse. The Matrix inverse method was used in solving the system of equations on SCILAB and Microsoft Excel and the answers obtained were all the same.

Question 2 was in two parts. Part A dealt with the determination of the temperature in a Resistance Temperature Detector (RTD) operation. Using the Newton-Raphson and Interval Bisection methods, the temperature of the detector when its resistance was 300Ω was calculated as 260.22°C. The Excel and SCILAB solutions also employed the Newton-Raphson equation to calculate the required temperature. In part B, the volume of a real gas was determined at different pressures by using the Van der Waals equation. The problem was solved in SCILAB with the "fsolve" function, and in Excel using "Goal Seek". Pressure was then plotted against volume and plotted in the Excel sheet. The plot showed an inverse relationship between pressure and volume (Boyle's Law).

A model was derived for a liquid tank system with inlet and outlet volumetric flow rates in question 3. The model related the volumetric inlet flow to the level of liquid in the tank. The model derived was a lumped parameter model. An expression for the dynamic response of the liquid level to change in operation commands was also determined by linearising the non-linear model equation. This was done using Taylor Series Expansion, and the resulting ODE was evaluated with Laplace Transforms. The solution of the resulting equation was taken to SCILAB to show the response of the system to different step lengths by using the Euler Explicit function and Runge-Kutta 4th order methods. The R-K graph is in reality more accurate than Euler's method since it more accurately accounts for the spatial and other differences in the system.

In question 4, the method of least-squares was used to best fit a function of the form $\ln(k) = C + b\ln(T) - D/T$ to the data given. This data showed the measurements of the rate coefficient, k for a reaction at different temperatures, T. By manually developing a system of three simultaneous equations using least squares and Crammer's rule, the values of C, b, and D were obtained. The values computed manually varied from the ones gotten in the software due to approximation and possible random errors. From comparing the function and constants gotten to the log of the Arrhenius equation, $k = AT^b e^{-E_a/RT}$, the values of A and E_a were deduced. All the problems were solved, typed, and formatted properly.

Things We Learned

- The MATLAB backslash operator (`A\b`) is the most numerically stable general-purpose method for solving linear systems.
- Newton-Raphson requires a good initial guess but converges much faster than bisection.
- RK4 offers a significant accuracy improvement over Euler's method at the same step size, with minimal additional computational cost.
- The least-squares design-matrix approach unifies single and multiple regression into one framework.
- Flash calculations are sensitive to the choice of K-value correlation; the Wilson equation provides a simple but reasonable first estimate.
- Back-substitution validation (residual check) is an essential step to verify any numerical solution.